APPENDIX 2

The Investograph Method of Appraising Security Values

Management is the most important single factor in a business. Poor management in a field which is most favored by general conditions usually results in disaster, while effective direction will frequently yield profits even under the handicap of severe competition and less favorable conditions.

In appraising character of management, it is results which count and not reputation. The record of past earnings trend, indicative of the results already accomplished for stockholders, provides an excellent basis for judging what the future may bring.

Yet it is frequently no simple matter to determine exactly the earnings trend of a particular company. Mergers and acquisitions, which are effected through the issuance of large blocks of stock, dilute the proportionate equity of the original stockholders. If profits of acquired companies are not sufficient to maintain per-share earnings on the stock issued for their purchase, then earnings on all shares are reduced. It will, therefore, be recognized that even substantial increases in total net income may be accompanied by an actual decline in earnings trend in so far as the individual stockholder is concerned.

The publishers of Investograph Service realized the vital necessity of furnishing earnings records properly adjusted for these factors and the desirability of providing a "picture" of past history so that the trend of earnings and other fundamentals could be shown accurately for clearer visualization.

In the accompanying Investograph charts of Eastman Kodak and du Pont, 15 years of historical statistics, all in their proper relationship, are revealed at a glance. The solid red line, labelled "Earnings on Original Share," reflects earnings on one share of stock, presumed to have been held from the beginning of the period charted. It is accurately adjusted for stock splits and dividends and shows the true earnings trend. Judging by earnings performance, du Pont has enjoyed an excellent record of increasing profits; Kodak a relatively poor one.

Earnings on original share for Kodak held close to a fixed level from 1919 through 1930 and, while it might be argued that this is favorable in the sense that earnings stability was shown, we should not lose sight of the fact that during this period general business enjoyed impressive growth and the motion picture industry was one of the leaders, with a resultant increase in the use of film. It is believed by many that Kodak should have profited substantially from this situation and that mere maintenance of earnings should be considered as unfavorable.

In comparison, the solid red line on the Investograph chart for du Pont makes an excellent showing. It is true that this company suffered more severely from an earnings standpoint in the depression of 1921, but a stockholder owning one share of du Pont in 1919, which for that year earned $13.30, saw his profits grow, assuming he held his many stock dividends, to a peak of $110.46 in 1929. Even in 1932, earnings were double the figure shown in 1919. Against this, earnings for Kodak in 1932, as compared with 1919, showed a decrease of 72%. Results in 1934, compared with 1919, show Kodak earnings as worse by 32% while du Pont is better of by 342%. If one accepts the theory that both companies enjoyed equal opportunities for growth, the evidence seems to indicate that du Pont has had the more successful management.

As the result of good management and a consequent substantial improvement in earnings, du Pont common, even at the low market price of the depression in 1932, sold at about 100% above the low price in 1922- after adjustment for stock dividends and splits. On the other hand, the price of Kodak stock, (adjusted for the 10 to 1 split in 1922), dropped considerably below its 1920 bottom.

The black-dashed line shows the number of shares of common stock outstanding at the end of each year. Where increases result from stock dividends or splits, this information is given. In the case of Kodak, there was one split of ten shares for one in 1922. With du Pont, however, splits and stock dividends have been more frequent and, without the adjustment provided which compensates for these — the Investograph "Earnings on Original Share" line and supporting figures — proper analysis of past performance would be complicated by the necessity of lengthy and involved computations.

The red-dashed line shows earnings as reported on the number of shares outstanding at the end of each year. The solid black line indicates cash dividends paid out of these earnings. One sees that with Kodak from 1923 and du Pont from 1926, it has been a corporate policy to distribute a large proportion of profits to stockholders — a policy which in all probability will be continued in future.

The black dot-dashed line shows the cost of carrying debts represented by bonds and debentures as well as divided requirements on any preferred stock issues outstanding. It is the sum of all fixed charges which must be paid before the proceeds may be applied to the common shares. By comparing the trend of this line with the line "Net for Common," it is possible to see in a moment whether capital position is getting better or worse. With du Pont, net income for the common shares was approximately twice requirements for debenture bonds in 1919. In 1934, however, this ratio had increased to 6 to 1. Prior charges with Kodak, represented by a relatively small preferred stock issue, have remained the same throughout the entire period and are inconsequential. It is worth noting, however, that Net Income of nearly $18,000,000 in 1919 stood at just a little more than $14,000,000 in 1934.

The small, green, supplementary chart near the bottom of each page gives a breakdown of yearly earn-

ings as reported either quarterly or semi-annually for a period of five years. This chart is especially valuable in those businesses where earnings are influenced by seasonal conditions, since they make it possible to analyze accurately the significance of current interim earnings as reported. In the lower section of this chart is a table in which earnings are adjusted backward on the basis of shares presently outstanding, thus making it possible to consider the trend of earnings in the light of the number of shares currently issued.

Monthly price ranges, reproduced by means of red vertical lines, have been found particularly useful in those cases where seasonal influences are effective. Loew's, Inc. normally can be bought in June and sold out at a profit in November or December; the tobacco stocks ordinarily reach a buying zone in March. Where seasonal characteristics have persisted over a long period of years, such information can be used advantageously in planning an investment or speculative program.

To support the charts, much interesting and important information is furnished on the back of the form. This includes a brief description and history of the business, comparative balance sheets for two years with each item shown in its percentage of total, a three year chart of income account, which materially simplifies consideration of this data, and a comparative statistical analysis of income statements and balance sheets.

The latter, exclusive with Investographs, is of great importance in determining upon the current strength or weakness of a given company and the conservatism, or lack of it, in accounting policy. This breakdown of income statements and balance sheets makes it possible to answer many questions including the following:

Is the profit margin on sales improving or shrinking?

What proportion of book value is represented by patents or goodwill?

Is the charge to depreciation excessive or inadequate?

Is the ratio of current assets to current liabilities satisfactory?

Is working capital sufficient or is it made up largely of inventories?

Are plant and equipment over or under valued?

Does overcapitalization indicate a dangerous situation?

Do interest and preferred dividend requirements constitute a source of probable trouble?

Are any bonds maturing within 3 years?

The figures are shown for three years so as to indicate trend and are also given on a per share basis for the current year. Percentages of increase or decrease are provided so that the full story is there and no further figuring is necessary. Companies in different lines of business are so analyzed as to present all the available characteristic data.

Much of this information is in ratio form. One may ask: "How can I properly interpret this?" As a part of the complete Service, there is a series of nine pages, called the "Investograph Comparative Analysis by Industries." Here, all the various companies are grouped according to classification of industry and the individual ratios as well as averages for each group are shown. Thus, it will be recognized that an accurate basis has been provided for proper appraisal of all factors. In addition, each Service is equipped with a series of instruction sheets which explain the interpretation and significance of all chart lines, ratios and other data.

Eastman Kodak and du Pont were selected as specimens in this instance because of the interesting comparison available in the chart forms. From a balance sheet standpoint, however, both companies are very strong so that a study of accounting policy, which is in each case conservative, and of current positions which are equally satisfactory, fails to disclose the usual value of the data provided here.

There is, however, an interesting story revealed in the comparative analysis of operating results achieved by U.S. Rubber and Firestone in 1934. The figures given below are taken from the "Comparative Statistical Analysis" tables appearing on the backs of the accompanying specimen Investographs. Each item has been assigned a reference number which is repeated in green on the forms in order to facilitate reference:

	U.S. Rubber		Firestone
1. Gross Sales	$105,476,872		$99,130,244
2. Sales Increase 1934 over 1933	19.4%		31.5%
3. Sales Per Share	$72.03		$52.24
4. Profit or Loss Per Share Loss	$3.15	Profit	$ .71
5. Book Value Per Share Less Intangibles	$7.83		$28.14
6. Depreciation Rate	4.6%		6.6%
7. Sales to Fixed Assets (Plant)	.805		1.13
8. Sales to Net Worth	1.49		1.86
9. Net Profit to Sales Loss	.044	Profit	.014
10. Bond Maturities Within 3 Years	10,198,883		None
Stock Prices July 25, 1935	13-1/2		15-1/4

It seems quite obvious from the foregoing that the management of Firestone was superior to that of U.S. Rubber. The latter enjoyed the greater gross sales, yet despite this and the fact that owned rubber plantations contributed a net profit of $1,735,842 (as covered in the last paragraph of the general description), U.S. Rubber ended the year with a deficit of well over $4,000,000 and a loss of 4.4 cents on every dollar's worth of business done. Firestone, in comparison, and without the benefit of income from rubber plantations, showed a net profit of $1,355,051 and earned 1.4 cents on every dollar of sales.

Attention is called to the fact that the deduction from income for depreciation by U.S. Rubber was 4.6%, while Firestone charged 2% more than this, or 6.6%. If the rate established by the latter had been on the same basis as U.S. Rubber, 2% of the item listed in the asset side of the Comparative Balance Sheet as "Land, Buildings & Machinery" would have increased earnings as reported from 71 cents to 80 cents per share. Likewise if the rate charged by U.S. Rubber were the same as established by Firestone, the loss would have been $3.30 per share instead of $3.15.

What is the average depreciation rate in the rubber business? The average for the industry in 1934 was 5.2% as will be seen by referring to the last line on page 1 of the "Investograph Comparative Analysis by Industries" under sub-title 9, "Depreciation ¢ Fixed Assets". On this same page under "Per Cent of Capitalization" the general balance sheet position of Firestone is seen to be considerably superior to that of U.S. Rubber. Of the former's total assets, nearly 44% represents common stock equity whereas the latter shows less than 5%. Moreover, Firestone's assets include only 1.3% for patents, deferred charges, etc., while these items for U.S. Rubber total 29.6%. The capital position of Firestone is considerably more conservative than the industry as a whole while that of U.S. Rubber compares unfavorably.

Now, by referring to the Investograph chart forms, we see further evidence of the superiority of Firestone. Since 1919 the company has shown a loss for only three years; namely, in 1921, 1930 and 1933. Against this, U.S. Rubber lost money in 1921; thereafter showed earnings through 1927, but has been in the red ever since. Since 1919 Firestone has failed to pay dividends on its common stock in only two years. U.S. Rubber has paid nothing to its shareholders since 1921. Charges prior to the common stock in the case of U.S. Rubber amount to nearly $9,000,000 annually while the figure for Firestone is less than one-half of this.

At the time this is written (July 25th, 1935), there is relatively little difference in the price of the two stocks. The evidence seems strongly to suggest that, comparatively, Firestone is under-priced.

One of the most interesting and practically useful features of Investograph Service is its monthly publication of Investograph Comparative Market Charts. Extending back through 1932, and maintained up-to-date by monthly cumulative supplements, these charts show the weekly market action of the various stocks covered, in direct comparison with the Dow Jones averages. The industrial, railroad or utility average is used depending upon the general classification of the company.

By means of these charts, market characteristics of the various stocks may be seen at a glance. Because a ratio scale is used, each chart is directly comparable with any other and, as a consequence, accurate comparisons can be drawn with reference to stock habits. One sees that certain issues regularly both gain and lose more, percentagewise, in advances and declines while others are habitually slow movers. These charts have been found very useful in the formulation of market programs.

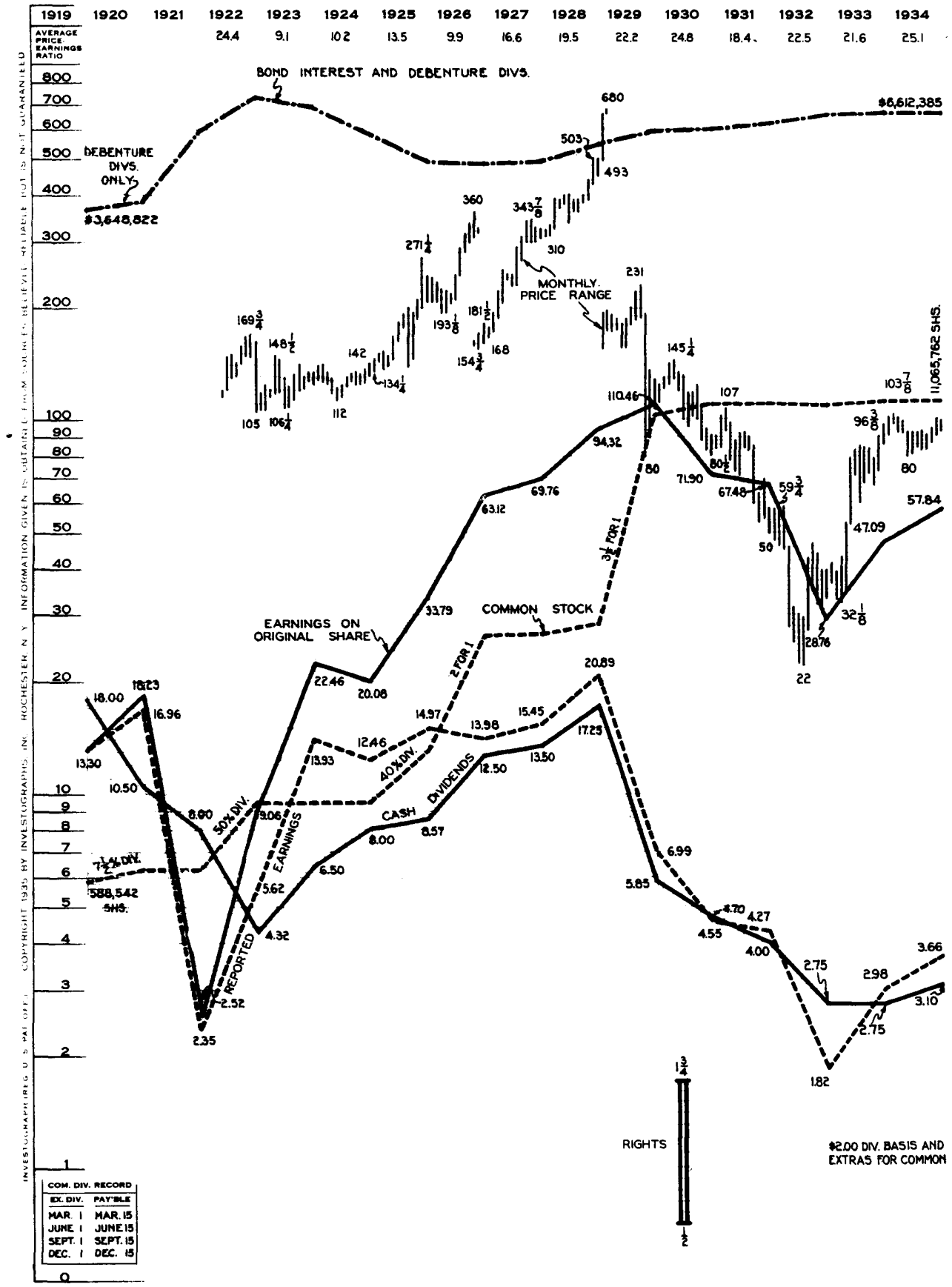
Investograph Service is kept up-to-date by means of weekly mailings, which include such new charts as are available; quarterly and semi-annual earnings of the various corporations, as published, so that these may be entered in the tables which are provided at the top of each chart sheet; and news items in the form of "Stickers" which are affixed to the charts affected. The Complete Investograph Service covers the 250 most active stocks as listed on the New York and Curb Exchanges and is published by Investographs, Inc., Rochester, N.Y.

DU PONT (E.I.) DE NEMOURS & CO (DD)

Listed N.Y.S.E. May 1922.
Dividends shown for 1922, 1925, 1926 and 1929 are on the basis of the stock after splits and dividends.
Earnings for 1930 to 1933 include undistributed equity of subsidiaries.

COMPARATIVE NET INCOME

		Total	Per Sh.		Total	Per Sh.
Quar.	3/31/34	$ 9,982,994.	$ .90	3/31/35	$ 9,372,733	$ .83
"	6/30/34	10,589,694.	.96	6/30/35	$ 9,847,276	$ .89
"	9/30/34	13,760,387.	1.24	9/30/35		
"	12/31/34	6,141,955.	.56	12/31/35		



DU PONT (E.I.) DE NEMOURS & CO.

One of the foremost chemical enterprises, this company is the largest manufacturer in the United States of powder and explosives. Other principal products manufactured, directly or through controlled or affiliated companies, are paints, lacquers, "Cellophane," rayon, fabrikoid, pyralin, motion picture film, industrial and ethyl alcohol, dyestuffs, ammonia, chemicals, acids, rubber coated fabrics, "Duco" and other related products.

As of December 31st, 1934, through its holdings in General Motors Securities Co., or by direct Investment, the Company owned a total of 10,000,000 shares of General Motors Corporation common stock, constituting 22.99% of all the outstanding common stock. These holdings were equivalent to approximately nine-tenths of one share of General Motors common stock for each share of common stock of Du Pont outstanding at the end of the year. In 1934 income from General Motors stock represented *approximately 37% of the earnings as reported.*

The company owns, or has a substantial interest in 28 companies, most of which are chemical enterprises. Extensive research laboratories are maintained for the improvement of present processes and products necessary to holding or improving the company's position in the chemical industry. During 1931 a new synthetic rubber called "DuPrene" was developed. This closely resembles natural rubber in strength, elasticity, toughness and other physical properties and in emergency is adaptable for any commercial purpose requiring the use of rubber.

During 1932 this company acquired majority interest in the Remington Arms Company consisting of 56% of the outstanding shares of common and in addition acquired 34,859 shares of 7% first preferred stock.

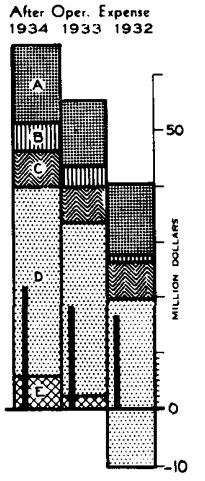
Audit by Price, Waterhouse & Co.

COMPARATIVE STATISTICAL ANALYSIS

	1934	Per Cent Change	1933	Per Cent Change	1932	Per Share 1934
Current Assets	$124,025,723.	− 4.4	$129,771,061.	+20.9	$107,325,753.	$11.21
Current Liabilities	19,155,627.	+15.4	16,601,263.	+56.2	10,630,857.	1.73
Cur. Assets ÷ Cur. Liabil.[1]	6.5		7.8		10.1	
Working Capital	104,870,096.	− 7.3	113,169,798.	+17.0	96,694,896.	9.48
Inventory	43,669,984.	+29.1	33,835,935.	+18.5	28,557,810.	3.95
Working Cap. ÷ Inventory[2]	2.4		3.3		3.4	
Sales	Not Avail.		Not Avail.		Not Avail.	
Net Profit for Common	40,475,030.	+22.9	32,921,253.	+66.5	19,769,395.	3.66
Net Worth	400,044,637.	+ 2.4	390,812,974.	− .2	391,505,764.	36.15
Patents, Goodwill, etc.	29,905,938.	+18.7	25,191,470.	....	25,193,820.	2.70
Net Profit ÷ Net Worth[3]	.101		.084		.050	
Net Profit ÷ Capitalization[4]	.079		.066		.039	
Depreciation	13,505,789.	+ 4.7	12,904,102.	− .8	13,009,753.	1.22
Dep. ÷ Fixed Assets[5]	.049		.052		.053	
Times Deb. Stk. Div. Earned	7.1		6.0		4.0	
Times Prior Chgs. Earned	7.1		6.0		4.0	
Bond Maturities Within 3 Yrs.	Not Avail.					

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Amount of profit from each dollar of net worth. (4) Amount of profit from each dollar of capitalization. (5) Amount charged to depreciation for each dollar of plant. *Note:* Increasing trend in ratios denotes improvement. Decreasing trend in ratios is unfavorable.

INCOME ACCOUNT



(A) Depreciation.
(B) Taxes.
(C) Deb. Dividend.
(D) Common Dividend.
(E) Surplus.
Other Income shown by inside vertical lines.

COMPARATIVE BALANCE SHEET

ASSETS

	Dec. 31, 1934	%	Dec. 31, 1933	%
Cash	$30,879,071.	4.9	$18,838,539.	3.1
Marketable Securities*	30,816,771.	4.9	58,010,388.	9.5
Accts. & Trade Accept. Rec.	18,659,897.	3.0	19,086,199.	3.2
Inventories	43,669,984.	6.9	33,835,935.	5.6
Total Current Assets	$124,025,723.	19.7	$129,771,061.	21.4
Plants & Properties	275,413,934.	43.6	246,724,457.	40.7
Gen. Motors Com. Stock**	157,000,000.	24.9	154,500,000.	25.5
Miscellaneous Securities	38,326,063.	6.1	41,331,026.	6.8
Notes Rec. for Empl. Stk.	5,259,694.	.8	7,146,576.	1.2
Patents, Goodwill, etc.	29,905,938.	4.7	25,191,470.	4.2
Deferred Items	1,188,929.	.2	966,474.	.2
	$631,120,281.	100.0	$605,631,064.	100.0
*Market Value	$31,019,829.		57,531,135.	

**10,000,000 shares at $15.70 in 1934 and $15.45 in 1933.

LIABILITIES

	Dec. 31, 1934	Dec. 31, 1933
Accounts Payable	$ 8,411,765.	$12,990,039.
Div. Pay. on Deb. Stk.	1,639,422.	1,639,926.
Deferred Items	9,104,440.	1,971,298.
Total Current Liabil.	$19,155,627.	$16,601,263.
Res. for Depreciation	75,540,299.	64,299,414.
Res. for Insurance, etc.	24,413,506.	23,453,963.
Prov. for Bonus Awards	1,536,412.	.....
Subsidiary Bonds	1,135,000.	1,135,000.
$6 Cum. Deb. Stk. ($100 par)	109,294,800.	109,328,450.
Common Stock ($20 par)	221,315,240.	220,467,740.
Surplus	178,729,397.	170,345,234.
	$631,120,281.	$605,631,064.

EARNINGS TREND BASED ON SALE OF "RIGHTS" AND PROCEEDS INVESTED IN ADDITIONAL STOCK

1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
13.30	18.23	2.53	9.06	22.46	20.09	33.79	63.12	69.76	94.32	110.46	72.49	68.03	29.00	47.48	58.31

Explanation: "Earnings on original share," as indicated by the solid red line on all Investographs are adjusted to compensate for stock dividends and stock splits only, "Rights" being disregarded and considered as income. Frequent or valuable offerings of "Rights," however, have a tendency to dilute equities and for this reason some consider that funds received from sale of "Rights" represent partial liquidation of invested capital. For this reason an earnings trend based upon sale of "Rights," and the investment of the proceeds of those "Rights" in additional stock, is given in the above table which, of course, is also adjusted for stock dividends and splits.

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed

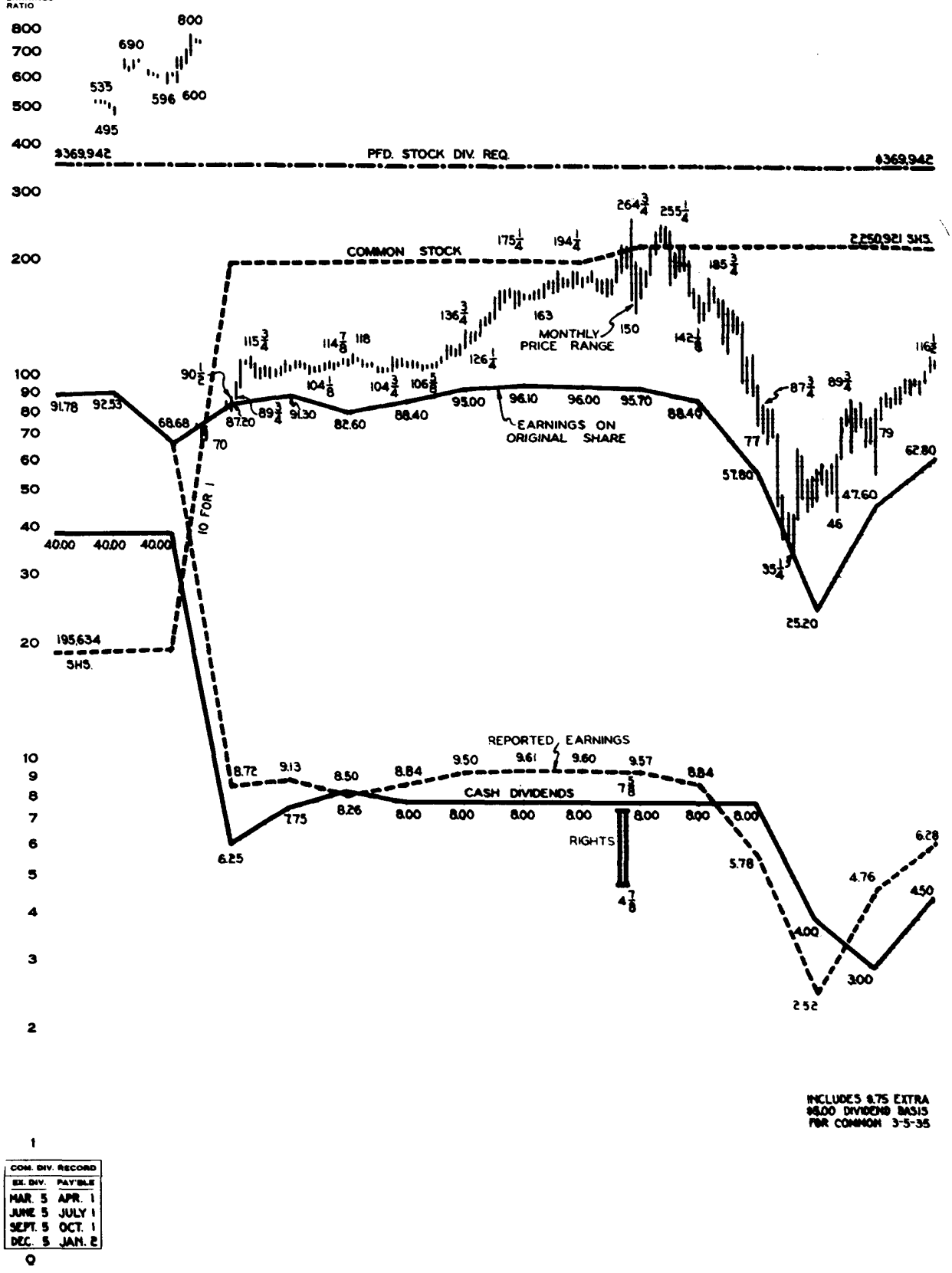
EASTMAN KODAK CO. (EK)

Dividends as shown in 1922 are on the basis of the new stock.
Adjustments on foreign exchange amounting to $1.67 per share in 1931 are charged to contingency reserve.
Net income for 1933 includes profit on foreign exchange adjustment in excess of amount credited to contingencies amounting to $.65 per share.

COMPARATIVE NET INCOME

	Total	Per Sh.		Total	Per Sh.
24 wks. end. 6/16/34	$6,574,934.	$2.91	6/16 35	$ 6,878,215	$3.06
28 wks. end. 12 29 35	7,558,371.	3.37	12 29 35		

	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
AVERAGE PRICE-EARNINGS RATIO		5.6	9.4	9.0	11.2	13.3	12.6	12.8	15.7	18.6	21.7	22.5	22.7	24.4	14.3	15.6



COM. DIV. RECORD	
EX. DIV.	PAYABLE
MAR. 5	APR. 1
JUNE 5	JULY 1
SEPT. 5	OCT. 1
DEC. 5	JAN. 2

Q

EASTMAN KODAK CO.

The largest manufacturer in the world of cameras, motion picture film and photographic supplies, this company has passed its fiftieth year. "Kodak" products and service are distributed through subsidiary companies or authorized dealers in all countries. The main plant of the Eastman Kodak Company is located at Rochester, N. Y., where during the past few years a broad program of plant expansion has been carried on including the acquisition of large manufacturing plants in France, Germany and the United States. In all, there are 15 plants; 6 in the United States, 3 in Germany, 2 in England and one each in Canada, France, Hungary and Australia.

In 1930 production of cellulose acetate for the manufacture of safety films and yarn for the rayon industry was begun on a large scale at the Tennessee-Eastman Corp. plant at Kingsport, Tenn., where more than 3,000 are now employed. The company, also, acquired the gelatine plants and business of the American Glue Company at Peabody, Mass., now known as the Eastman Gelatine Corp. The Recordak Corp., a subsidiary, markets a device for photographing checks which is used in the banking field. Eastman Teaching Films, Inc., has built up a large library of educational films for teaching work.

In March, 1931, the company announced Kodak Verichrome Film, considered to represent the greatest advance in snapshot photography since non-curling film was introduced in 1903.

In 1932 the Cine-Kodak Eight for amateur motion pictures was introduced followed in 1933 by a 16mm. Ciné-Kodak Special for the use of advanced amateurs and scientists.

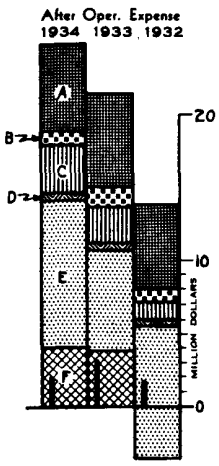
Audit by Price, Waterhouse Co.

COMPARATIVE STATISTICAL ANALYSIS

	1934	Per Cent Change	1933	Per Cent Change	1932	Per Share 1934
Current Assets	$91,522,874.	+14.6	$79,884,343.	+21.7	$65,666,575.	$40.66
Current Liabilities	15,109,871.	+38.3	10,926,860.	+38.7	7,878,565.	6.71
Cur. Assets ÷ Cur. Liabil.[1]	6.1		7.3		8.3	
Working Capital	76,413,003.	+10.8	68,957,483.	+19.3	57,788,010.	33.95
Inventory	39,076,080.	+15.3	33,904,121.	+15.0	29,476,937.	17.36
Working Cap. ÷ Inventory[2]	2.0		2.0		2.0	
Sales	Not Avail.		Not Avail.		Not Avail.	
Net Profit for Common	14,133,305.	+31.5	10,749,102.	+89.0	5,688,807.	6.28
Net Worth	131,727,174.	+ 3.0	127,937,751.	+ 3.2	123,941,412.	58.52
Patents, Goodwill, etc.	None		None		None	
Net Profit ÷ Net Worth[3]	.107		.084		.046	
Net Profit ÷ Capitalization[4]	.102		.080		.044	
Depreciation	6,022,174.	+ 3.5	5,818,051.	+ 1.0	5,757,626.	2.68
Dep. ÷ Fixed Assets[5]	.052		.052		.054	
Times Pfd. Div. Earned	39.2		30.1		16.4	
No Bonds Outstanding						

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Amount of profit from each dollar of net worth. (4) Amount of profit from each dollar of capitalization. (5) Amount charged to depreciation for each dollar of plant. *Note:* Increasing trend in ratios denotes improvement. Decreasing trend in ratios is unfavorable.

INCOME ACCOUNT



(A) Depreciation.
(B) Other Charges.
(C) Taxes.
(D) Preferred Dividend.
(E) Common Dividend.
(F) Surplus.
Other Income shown by inside vertical lines.

COMPARATIVE BALANCE SHEET

ASSETS

	Dec. 29, 1934	%	Dec. 30, 1933	%
Cash	$14,683,443.	6.7	$17,276,786.	8.6
Marketable Securities	19,617,729.	9.0	12,338,873.	6.2
Accts. & Bills Receivable	18,145,622.	8.5	16,364,563.	8.2
Inventories	39,076,080.	18.0	33,904,121.	16.8
Total Current Assets	$ 91,522,874.	42.2	$ 79,884,343.	39.8
Land, Bldgs., Plant & Mach.	116,896,139.	53.8	111,155,834.	55.5
Invest. Affil. Companies	1,508,032.	.7	1,621,949.	.8
Other Investments	3,772,661.	1.7	4,612,861.	2.3
Kodak Empl. Assoc.	2,349,622.	1.1	2,454,521.	1.2
Deferred Charges	1,064,568.	.5	771,645.	.4
	$217,113,896.	100.0	$200,501,153.	100.0

LIABILITIES

	Dec. 29, 1934	Dec. 30, 1933
Accounts Payable	$11,078,273.	$ 9,146,184.
Dividends Payable	4,031,598.	1,780,676.
Total Current Liabil.	$15,109,871.	$10,926,860.
Res. for Depreciation	51,483,591.	46,346,455.
Gen'l. & Contingent Res.	12,627,560.	9,124,387.
6% Cum. Pfd. Stk. ($100 par)	6,165,700.	6,165,700.
Common Stock (no par)	22,509,210.	22,559,210.
Paid-in-Surplus	28,617,862.	28,782,600.
Earned Surplus	80,600,102.	76,595,941.
	$217,113,896.	$200,501,153.

EARNINGS TREND BASED ON SALE OF "RIGHTS" AND PROCEEDS INVESTED IN ADDITIONAL STOCK

1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
91.78	92.53	68.68	87.20	91.30	82.60	88.40	95.00	96.10	96.00	96.01	88.68	57.98	25.28	47.75	63.00

Explanation: "Earnings on original share," as indicated by the solid red line on all Investographs are adjusted to compensate for stock dividends and stock splits only, "Rights" being disregarded and considered as income. Frequent or valuable offerings of "Rights," however, have a tendency to dilute equities and for this reason some consider that funds received from sale of "Rights" represent partial liquidation of invested capital. For this reason an earnings trend based upon sale of "Rights," and the investment of the proceeds of those "Rights" in additional stock, is given in the above table which, of course, is also adjusted for stock dividends and splits.

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed

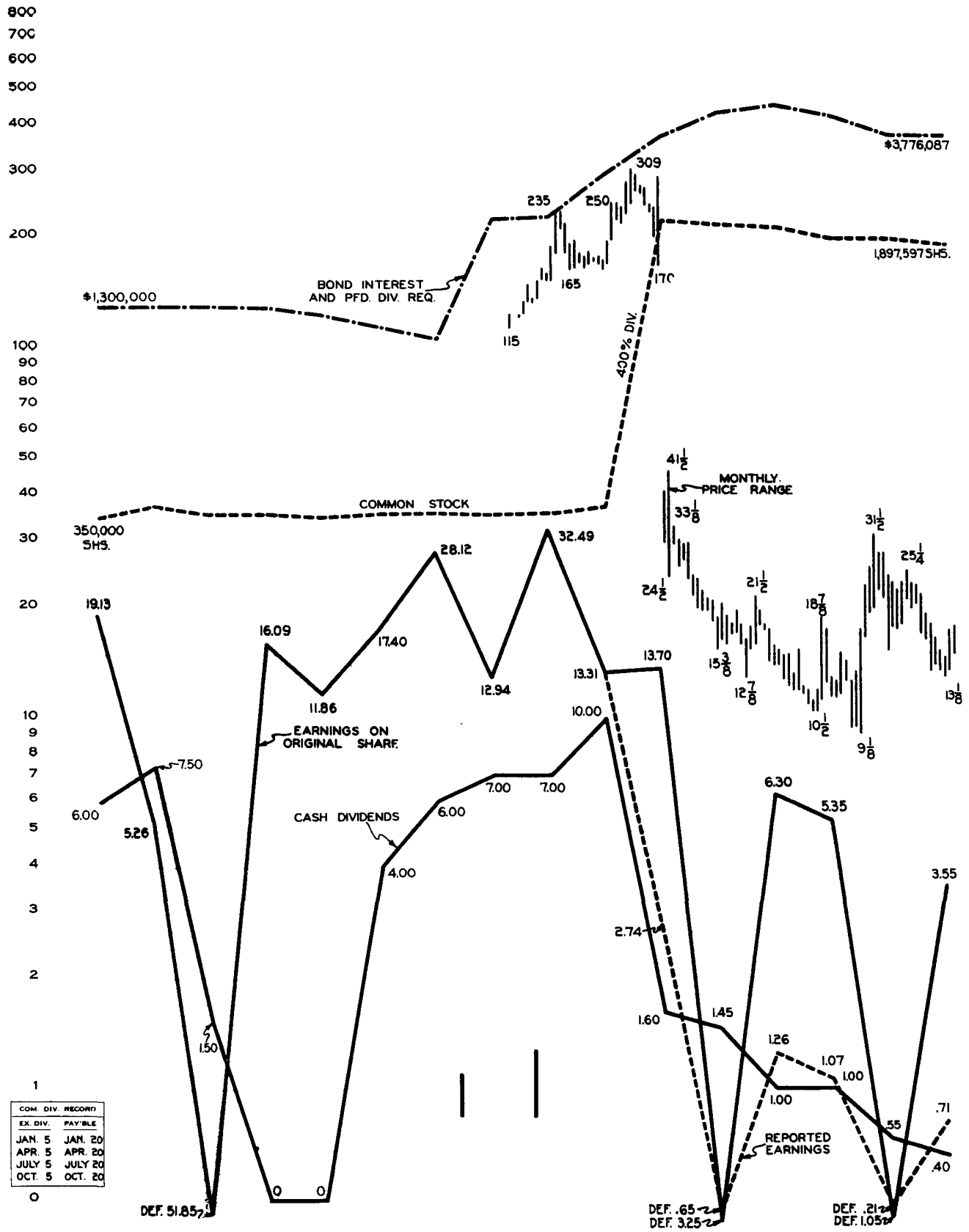
FIRESTONE TIRE & RUBBER CO. (FIR)

COMPARATIVE NET INCOME

Price range to 1926 Cleveland S. E. New York Curb to December 1929, N. Y. S. E. thereafter. Fiscal years end October 31st. Price earnings ratios and monthly price ranges are on calendar year basis, all other information reflects operations for the fiscal years ending October 31st and this date is plotted to correspond with this date on the chart form. Price of rights in October 1929 is not available. Increase in capital stock, except as shown, is due to sales for cash. Net income for 1931 is before adjustment of current assets for foreign subsidiaries equivalent to $.86 per share.

		Total	Per Sh.		Total	Per Sh.
Quar.	1/31/34			1/31/35		
"	4/30/34	$†122,254.	$†.06	4/30/35	$ 757,158	$ .40
"	7/31/34			7/31/35		
†6 mo. "	10/31/34	†1 232,797.	†.65	10/31/35		

	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
AVERAGE PRICE-EARNINGS RATIO	9.0	25.7	DEF.	4.0	6.4	4.8	4.3	8.7	5.2	15.6	15.7	DEF.	13.6	13.7	DEF.	27.0



COM. DIV. RECORD	
EX. DIV.	PAY'BLE
JAN. 5	JAN. 20
APR. 5	APR. 20
JULY 5	JULY 20
OCT. 5	OCT. 20

FIRESTONE TIRE & RUBBER CO.

One of the largest manufacturers of rubber tires in the world, this company either directly or through subsidiaries also manufactures rubber boots, shoes and heels, electric storage batteries, spark plugs, brake lining, automobile rims and other automobile accessories. A large percentage of the manufacturers of automobiles and trucks use Firestone tires as original equipment.

The company's products are distributed through approximately 200 branches and warehouses to 30,000 dealers in this country and through about 300 branches and distributing agencies in foreign countries.

In order to maintain and build up markets the company since 1929 has been developing a chain of retail service stores which are reported by the President, Mr. Harvey S. Firestone, to have made a profit for the first time in 1934. Nearly 500 stores are now operated. It was also stated that export business and foreign factories in Canada, England and the Argentine operated on a profitable basis and their new factory in Spain is showing promise.

Mr. Firestone informed his stockholders that the company had enough rubber and cotton on hand below the market to t[illegible] of requirements for 1935.

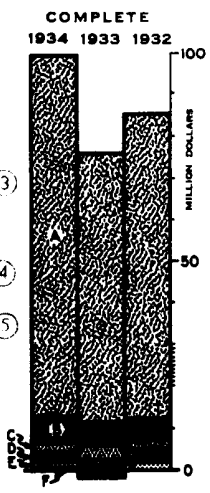
Audit by Lybrand, Ross Bros. & Montgomery

COMPARATIVE STATISTICAL ANALYSIS

	1934	Per Cent Change	1933	Per Cent Change	1932	Per Share 1934
Current Assets	$62,087,151.	+ 9.4	$56,766,302.	+ 6.6	$53,262,247.	$32.72
Current Liabilities	14,302,315.	+117.9	6,563,145.	+67.3	3,923,386.	7.54
Cur. Assets ÷ Cur. Liabil.[1]	4.3		8.6		13.6	
Working Capital	$47,784,836.	− 4.8	50,203,157.	+ 1.8	49,338,861.	25.18
Inventory	36,326,223.	+ 27.3	28,541,261.	+ 6.4	26,826,143.	19.14
Working Cap. ÷ Inventory[2]	1.3		1.8		1.8	
Sales	99,130,244.	+ 31.5	75,402,268.	− 10.6	84,337,173.	52.24
Sales ÷ Inventory[3]	2.7		2.6		3.1	
Sales ÷ Net Worth[4]	1.86		1.41		1.51	
Sales ÷ Fixed Assets[5]	1.13		.895		.987	
Net Profit for Common	1,355,051.	....	*def. 417,906.*	....	2,127,746.	.71
Net Profit ÷ Sales[6]	.014		*def. .006*		.025	
Net Worth	53,405,717.	− .2	53,523,815.	− 4.2	55,858,328.	28.14
Patents, Goodwill, etc.	None		None		None	
Net Profit ÷ Net Worth[7]	.025		*def. .008*		.038	
Net Profit ÷ Capitalization[8]	.011		*def. .003*		.017	
Depreciation	5,804,295.	+ 3.2	5,625,097.	+ 7.9	5,211,893.	3.06
Dep. ÷ Fixed Assets[9]	.066		.067		.061	
Times Subs. Bond Int. Earned	5.3		3.4		5.3	
Times Pfd. Div. Earned	1.4		.9		1.5	
Times Prior Chgs. Earned	1.4		.9		1.5	
Bond Maturities Within 3 Yrs.	None					

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Shows relation of annual sales to inventory. (4) Amount in sales for each dollar of net worth. (5) Amount in sales for each dollar of plant or productive assets. (6) Amount of profit on each dollar of sales. (7) Amount of profit on each dollar of net worth. (8) Amount of profit from each dollar of capitalization. (9) Amount charged to depreciation for each dollar of plant. *Note:* Increasing trend in ratios denotes improvement. Decreasing trend in ratios is unfavorable.

INCOME ACCOUNT



COMPARATIVE BALANCE SHEET

ASSETS

	Oct. 31, 1934	%	Oct. 31, 1933	%
Cash	$10,614,594.	6.6	$14,320,030.	9.2
Accts. & Notes Rec.	15,146,334.	9.4	13,905,011.	9.0
Inventories	36,326,223.	22.4	28,541,261.	18.4
Total Current Assets	$62,087,151.	38.4	$56,766,302.	36.6
Land, Bldgs. & Mchy.	87,627,546.	54.2	84,259,309.	54.4
The Firestone Park Land Co.	1,856,513.	1.1	1,909,867.	1.2
Employees, Stock Contracts	3,778,438.	2.3	2,748,013.	1.8
Stocks and Bonds (At Cost)	2,708,531.	1.7	2,353,330.	1.5
Misc. Accts. & Advs.	2,288,447.	1.4	2,135,386.	1.4
Invest. of Subs. Co.	.....	....	3,044,385.	2.0
Prepaid Int., Ins. & Taxes	338,720.	.2	275,295.	.2
Bond Disc. & Deferred Exp.	1,189,909.	.7	1,347,222.	.9
	$161,875,255.	100.0	$154,839,109.	100.0

LIABILITIES

	Oct. 31, 1934	Oct. 31, 1933
Notes Payable Banks	$ 5,000,000.	.....
Acct's Payable	5,895,590.	$ 3,842,485.
Accr. Taxes & Interest	3,406,725.	2,720,660.
Total Current Liabil.	$14,302,315.	$ 6,563,145.
Res. for Depreciation	27,984,482.	25,313,897.
Res. for Investments, etc.	875,000.	2,200,000.
Res. for Insurance	600,000.	600,000.
Subs. Funded Debt	17,350,000.*	18,962,800.
Minority Interest	760,241.	985,252.
6% Cum. Preferred Stk. ($100 Par)	46,597,500.	46,690,200.
Common Stock ($10 Par)	18,975,970.	19,708,490.
Earned Surplus	32,423,855.	31,841,392.
Paid-in-Surplus	2,005,892.	1,973,933.
	$161,875,255.	$154,839,109.

*Sinking Fund Requirements within one year $1,350,000

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed

U. S. RUBBER CO. (RU)

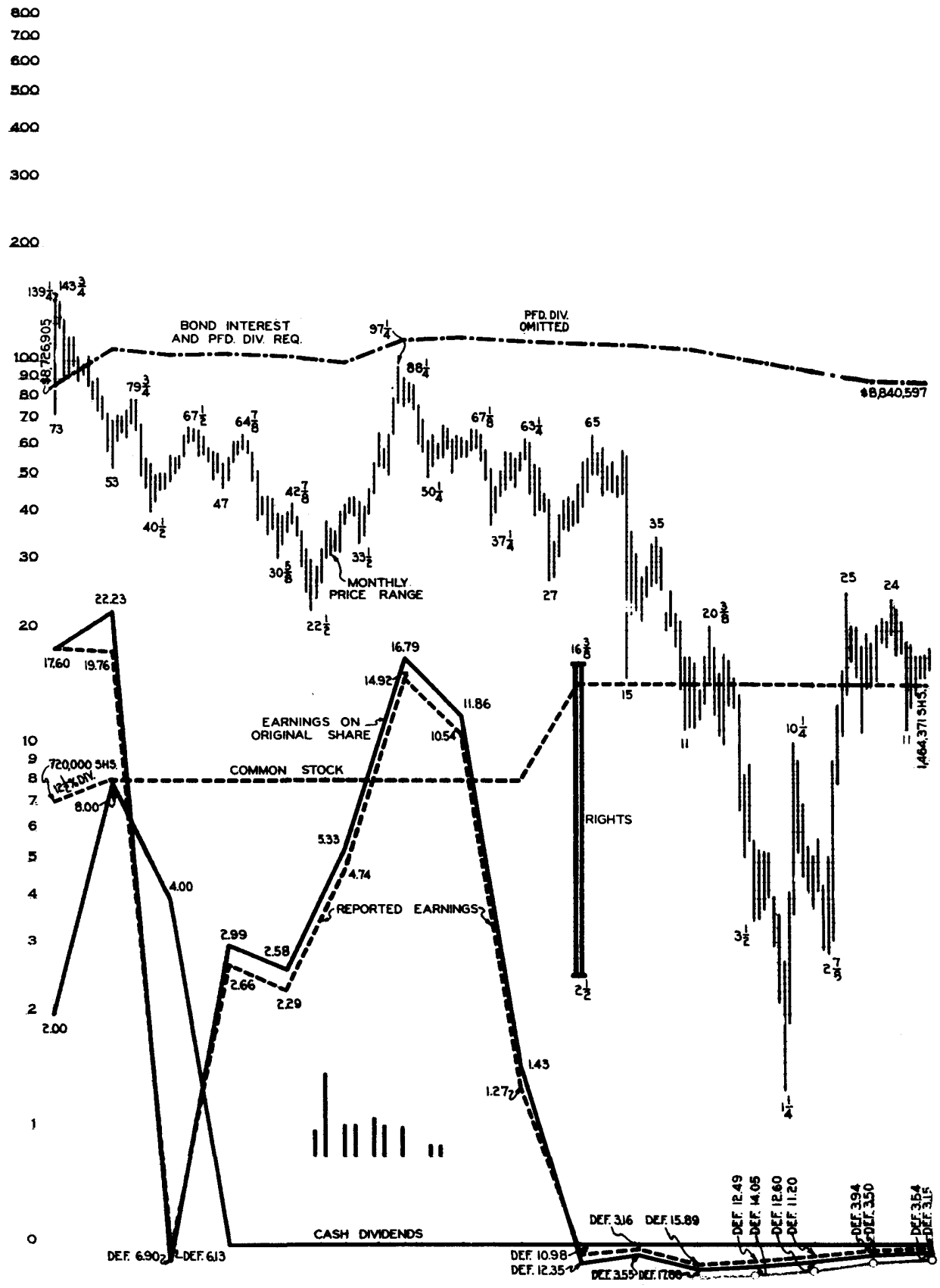
Earnings as shown for 1928-1930 are after inventory adjustments and for 1931 include inventory adjustment, foreign exchange adjustment and loss on securities equivalent to $2.75 per share.

Deficit for 1932 includes: adjustment of inventories ($1,108,386), foreign exchange ($366,203) and securities ($99,040) in all equal to $1.07 per share. 1933 earnings do not include loss on extraordinary charges of $1,679,713. and profit and bonds purchased of $996,462. in all equal to a loss of $.47 per share.

COMPARATIVE NET INCOME

†6 Mo. ending		Total	Per Sh.		Total	Per Sh.
Quar.	3/31/34			3/31/35		
"	6/30/34	$†d 2,577,776.	$†d 1.76	6/30/35	def. $1,945,784	d $1.33
"	9/30/34			9/30/35		
"	12/31/34	†d 2,035,617.	†d 1.39	12/31/35		

	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
AVERAGE PRICE-EARNINGS RATIO		5.0	DEF.	21.5	20.9	6.9	4.4	6.6	41.4	DEF.	DEF.	DEF.	DEF.	DEF.	DEF.	DEF.



The second largest manufacturer of rubber products, this company has in recent years been following a program of centralizing its production in Detroit. During 1930 this resulted in the closing of 11 factories and the sale of 4. Early in 1931 an interest was acquired in Samson Tire and Rubber Corp. of Los Angeles and the Gillette Rubber Co. of Eau Claire, Wis.

Tire casings and tubes represent the main activities of the company although a substantial business is done in rubber foot wear, "Uskide" soles, household goods, rain and sport coats, luggage, flooring, sporting goods, rubber industrial goods, chemicals and imitation leather and waterproof materials. The company is said to be the chief source of tire supply for Montgomery Ward & Co.

Through United States Rubber Plantations, Inc., the company operates the largest rubber plantations in the world, located in Sumatra and Malaya. The plantations showed a profit during 1934, after depreciation of $1,735,842.

Audit by Haskins & Sells

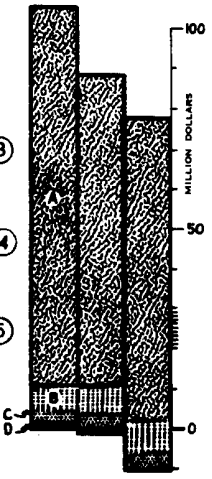
COMPARATIVE STATISTICAL ANALYSIS

	1934	Per Cent Change	1933	Per Cent Change	1932	Per Share 1934
Current Assets	$57,743,651.	+15.3	$50,099,681.	+ 3.9	$48,228,364.	$39.43
Current Liabilities	14,178,463.	+31.5	10,778,827.	−31.5	15,742,559.	9.68
Cur. Assets ÷ Cur. Liabil.[1]	4.1		4.6		3.1	
Working Capital	43,565,188.	+10.8	39,320,854.	+21.0	32,485,805.	29.75
Inventory	29,298,072.	+16.0	25,253,844.	+27.5	19,800,067.	20.01
Working Cap. ÷ Inventory[2]	1.5		1.6		1.6	
Sales	105,476,872.	+19.4	88,326,666.	+12.8	78,300,091.	72.03
Sales ÷ Inventory[3]	3.6	(1)	3.5	(2)	4.0	(3)
Sales ÷ Net Worth[4]	(8) 1.49		1.24		1.09	
Sales ÷ Fixed Assets[5]	.805 (7)		.674		.586	
Net Profit for Common	def. 4,613,393.	....	def. 5,131,815.	....	def. 16,400,003.	def. 3.15 (4)
Net Profit ÷ Sales[6]	def. .044 (9)		def. .058		def. .209	
Net Worth	70,633,830.	− .8	71,177,439.	− .8	71,783,776.	48.24
Patents, Goodwill, etc.	59,168,645.	+ .4	58,925,372.	....	58,925,372.	40.41
Net Profit ÷ Net Worth[7]	def. .065		def. .072		def. .228	7.83 (5)
Net Profit ÷ Capitalization[8]	def. .023		def. .025		def. .080	
Depreciation	5,966,400.	− 7.7	6,462,613.	− 1.2	6,541,313.	4.07
Dep. ÷ Fixed Assets[9]	.046 (6)		.049		.049	
Times Bond Int. Earned	1.2		1.0		Not Earned	
Times Pfd. Div. Earned	.5		.4		Not Earned	
Times Prior Chgs. Earned	.5 (10)		.4		Not Earned	
Bond Maturities Within 3 Yrs.	10,198,883.	($5,000,000—3 Yr. 6% Gold Notes due 6/1/36, $5,062,000—6½% Serial Gold Notes due 1935-6-7)				

INCOME ACCOUNT

COMPLETE

1934 1933 1932



(A) Oper. Expense.
(B) Depreciation.
(C) Interest & Misc. Chgs.
(D) Surplus.

* Non-Cum. Pfd. Div. of $5,208,728 not paid in 1934; $5,208,728 in 1933; $5,208,728 in 1932.

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Shows relation of annual sales to inventory. (4) Amount in sales for each dollar of net worth. (5) Amount in sales for each dollar of plant or productive assets. (6) Amount of profit on each dollar of sales. (7) Amount of profit on each dollar of net worth. (8) Amount of profit on each dollar of capitalization. (9) Amount charged to depreciation for each dollar of plant. *Note:* Increasing trend in ratios denotes improvement. Decreasing trend in ratios is unfavorable.

COMPARATIVE BALANCE SHEET

ASSETS

	Dec. 31, 1934	%	Dec. 31, 1933	%
Cash	$10,979,722.	5.0	$ 8,054,292.	3.7
Marketable Securities	253,754.	.1	139,002.	....
Accts. & Notes Rec.	17,212,103.	7.8	16,652,543.	7.7
Inventories	29,298,072.	13.3	25,253,844.	11.8
Total Current Assets	$ 57,743,651.	26.2	$ 50,099,681.	23.2
Properties, Plant & Equipt.	131,063,257.	59.3	130,958,245.	60.6
U. S. Rubber Plantations	24,108,837.	10.9	27,084,871.	12.5
Insurance Fund	1,071,628.	.5	1,034,632.	.5
Securities of Controlled Cos.	3,675,950.	1.7	3,661,575.	1.7
Other Securities	1,566,219.	.7	1,632,023.	.8
Goodwill, Patents, etc.	*		*	
Deferred Assets	1,490,791.	.7	1,622,830.	.7
	$220,720,333.	100.0	$216,093,857.	100.0

LIABILITIES

	Dec. 31, 1934	Dec. 31, 1933
Accts. & Accept. Pay.	$ 8,352,533.	$ 5,497,831.
Accr. Int. & Other Liabil.	4,340,930.	3,923,996.
Serial Notes Due	1,485,000.	1,357,000.
Total Current Liabil.	$14,178,463.	$10,778,827.
Res. for Depreciation	57,316,417.	53,371,673.
Other Reserves	3,377,585.	3,043,575.
Minority Interest	315,700.	338,700.
Funded Debt	68,957,883.	71,199,915.
8% Non-Cum. Pfd. Stk. ($100 par)	65,109,100.	65,109,100.
Common Stk. (no par) and Surplus*	11,465,185.	12,252,067.
	$220,720,333.	$216,093,857.

*Goodwill, Patents, etc., are carried on the books of the company at $59,168,645 and are shown in this report as a deduction from common stock and surplus to coincide with the balance sheet as submitted to the stockholders in previous reports. Because Goodwill and Patents are still carried, however, on the control records, this item is included in the statistical analysis figure for net worth.

EARNINGS TREND BASED ON SALE OF "RIGHTS" AND PROCEEDS INVESTED IN ADDITIONAL STOCK

1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930	1931	1932	1933	1934
17.60	22.23	d6.90	2.99	2.58	5.33	16.79	11.86	1.43	d13.39	d3.85	d19.38	d15.23	d13.66	d4.27	d3.84

Explanation: "Earnings on original share," as indicated by the solid red line on all Investographs are adjusted to compensate for stock dividends and stock splits only, "Rights" being disregarded and considered as income. Frequent or valuable offerings of "Rights," however, have a tendency to dilute equities and for this reason some consider that funds received from sale of "Rights" represent partial liquidation of invested capital. For this reason an earnings trend based upon sale of "Rights," and the investment of the proceeds of those "Rights" in additional stock, is given in the above table which, of course, is also adjusted for stock dividends and splits.

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed

INVESTOGRAPH COMPARATIVE ANALYSIS BY INDUSTRIES

The following tables facilitate comparison of income statement and balance sheet ratios between individual stocks as well as "averages" in each of the main classifications of industry.

The "averages" as shown are not merely averages of the ratio figures because such procedure would give as much "weight" to small companies as to large ones which, obviously, would not be proper. Instead, averages are determined by first adding together all the basic figures necessary before making final computations, thus automatically adjusting for relative size of the various companies in each group and giving a true cross-section of the different industries. The figures in heavy type are intended to call attention to favorable ratios, being those above the average.

Unless otherwise indicated by a date following company names, all computations are based upon income statements and balance sheets as of Dec. 31, 1934. The data furnished here includes only the most important ratios and figures and reference should be made to individual Investographs for complete historical and analytical information. Because the following tables require inclusion of all possible company reports, preparation of a new series will have to wait upon receipt of late final statements which will probably be in July 1936.

AGRICULTURAL IMPLEMENTS

See Foot-note Explanation	1	2	3	4	5	6	7	8	9		Per Cent of Capitalization			
NAME OF COMPANY	Cur. Assets ÷ Cur. Liabil.	Working Cap. ÷ Inventory	Sales ÷ Inventory	Sales ÷ Net Worth	Sales ÷ Fixed Assets	Net Profit ÷ Sales	Net Profit ÷ Net Worth	Net Profit ÷ Capitalization	Depreciation ÷ Fixed Assets	Times Prior Chgs. Earned	Common & Sur. less Intangibles	Deferred Chgs. Patents, etc.	Preferred & Minority Int.	Funded Debt
Case (J. I.) Co.	**28.6**	**2.6**	*	*	*	*	d.055	d.039	.023	**	68.4	3.3	28.3x	–0–
Caterpillar Tractor	**12.7**	1.9	2.3	.651	.911	.154	**.100**	**.100**	**.069**	**None**	99.1	.1	–0–	–0–
Deere & Co. (10/31/34)	**15.7**	**3.2**	*	*	*	*	d.061	d.030	.037	.2	48.1	.6	51.3x	–0–
International Harvester	10.1	2.2	*	*	*	*	d.008	d.006	.036	.7	71.9	.1	28.0x	–0–
Average	11.5	2.3	—	—	—	—	d.004	d.003	.038	.9	70.6	.5	28.9	–0–

AUTOMOBILES AND TRUCKS

Auburn Automobile (11/30/34)	**8.8**	1.6	3.2	1.15	.954	d.353	d.404	d.317	.044	**None**	77.9	.7	14.9y	6.5
Chrysler Corp.	2.3	1.3	**9.7**	**4.26**	**2.81**	.026	**.112**	**.083**	**.096**	4.4	72.0	1.8	–0–	26.2
Ford Motor of Canada	**12.9**	**7.6**	*	*	*	*	.064	.064	.053	**None**	97.5	2.5	–0–	–0–
General Motors	**4.1**	2.0	6.1	1.20	1.56	**.101**	**.121**	**.096**	.060	**10.3**	71.4	7.4	21.2	–0–
Hudson Motor Car	1.3	.5	**11.5**	**2.19**	1.06	d.062	d.135	d.135	.040	*	98.3	1.7	–0–	–0–
Mack Trucks	**13.1**	**2.5**	2.2	.442	.524	.001	.0004	.0004	.022	**None**	93.5	6.5	–0–	–0–
Nash Motors (11/30/34)	**21.4**	**13.4**	**9.2**	.573	**1.70**	d.083	d.047	d.047	**.071**	**None**	99.9	.1	–0–	–0–
Packard Motor Car	**4.0**	**2.9**	3.1	.351	.324	d.499	d.175	d.175	.041	**None**	99.0	1.0	–0–	–0–
Average	3.9	2.0	6.6	1.40	1.61	.060	.084	.068	.062	7.7	75.2	6.1	16.1	2.6

AUTOMOBILE PARTS AND ACCESSORIES

Bendix Aviation	**7.3**	**2.9**	*	*	*	*	.077	.077	**.066**	**103.1**	97.6	.8	.6	1.0
Bohn Aluminum	**6.9**	1.1	*	*	*	*	**.216**	**.201**	.052	**22.3**	90.8	2.6	–0–	6.6
Borg-Warner	4.3	1.9	*	*	*	*	**.135**	**.116**	**.063**	12.9	82.8	2.6	11.1	3.5
Briggs Manufacturing	3.1	2.2	*	*	*	*	**.195**	**.195**	.045	**None**	99.6	.4	–0–	–0–
Eaton Manufacturing	4.2	2.1	*	*	*	*	.087	.078	**.062**	**None**	85.6	3.9	10.5y	–0–
Electric Auto Lite	3.7	1.9	*	*	*	*	.068	.050	.036	3.9	71.9	1.9	25.1	1.1
Electric Storage Battery	**17.7**	**5.4**	4.6	.547	.740	.110	.060	.060	*	**750.1**	98.2	1.7	.1x	–0–
Motor Products	5.4	**3.4**	*	*	*	*	.020	.020	.036	**None**	98.5	1.5	–0–	–0–
Murray Corp.	2.8	1.3	*	*	*	*	d.072	d.061	.047	**	82.5	2.4	1.4x	13.7
Timken Roller Bearing	**15.6**	**3.2**	*	*	*	*	.085	.085	.052	**None**	99.5	.5	–0–	–0–
Trico Products	**6.5**	**12.1**	*	*	*	*	**.226**	**.226**	**.062**	**None**	95.5	4.5	–0–	–0–
Average	5.7	2.6	—	—	—	—	.098	.092	.052	21.0	92.3	1.7	4.3	1.7

AUTOMOBILE TIRES, RUBBER GOODS, ETC.

Firestone Tire (10/31/35)	4.3	1.3	2.7	**1.86**	**1.13**	**.014**	**.025**	**.011**	**.066**	**1.4**	43.9	1.3	40.1	14.7
Goodrich	6.1	1.4	**3.0**	1.09	**1.29**	**.005**	**.005**	**.003**	.049	**1.1**	22.1	36.1	18.4	23.4
Goodyear Tire	**11.8**	**1.7**	2.5	**3.98**	.849	**d.007**	d.029	d.006	.051	.9	17.9	1.6	48.8	31.7
U. S. Rubber	4.1	1.5	**3.6**	1.49	.805	d.044	d.065	d.023	.046	.5	4.9	29.6	31.9	33.6
Average	6.3	1.6	2.9	1.76	.967	d.008	d.015	d.006	.052	.9	19.5	18.7	34.5	27.3

*not available **not earned d—deficit x—no minority interest y—no preferred

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Shows relation of annual sales to inventory. (4) Amount in sales for each dollar of net worth. (5) Amount in sales for each dollar of plant or productive assets. (6) Amount of profit on each dollar of sales. (7) Amount of profit from each dollar of net worth. (8) Amount of profit from each dollar of capitalization. (9) Amount charged to depreciation for each dollar of plant.

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed.

AVIATION

See Foot-note Explanation	1	2	3	4	5	6	7	8	9		Per Cent of Capitalization			
NAME OF COMPANY	Cur. Assets ÷ Cur. Liabil.	Working Cap. ÷ Inventory	Sales ÷ Inventory	Sales ÷ Net Worth	Sales ÷ Fixed Assets	Net Profit ÷ Sales	Net Profit ÷ Net Worth	Net Profit ÷ Capitalization	Depreciation ÷ Fixed Assets	Times Prior Chgs. Earned	Common & Sur. less Intangibles	Deferred Chgs. Patents, etc.	Preferred & Minority Int.	Funded Debt
Aviation Corp.	3.7	**8.1**	*	*	*	*	d.177	d.173	**.272**	**None**	92.6	5.2	1.9y	.3
Curtiss-Wright	**5.1**	1.9	**6.0**	.567	.990	d.135	**d.076**	d.071	.041	**.2**	92.0	.6	6.2	1.2
Douglas Aircraft	**6.1**	1.3	2.4	**1.21**	**5.10**	**.007**	**.009**	**.009**	.046	**None**	97.8	2.2	–0–	–0–
Average	4.5	2.5	4.2	.661	1.26	d.096	d.097	d.043	.088	**	92.8	2.2	4.3	.7

BUILDING (EQUIPMENT, SUPPLIES)

American Radiator	9.9	2.1	2.6	.552	**.509**	**.017**	.009	.008	.032	1.7	83.4	1.0	7.9	7.7
Congoleum-Nairn	**20.5**	**2.7**	*	*	*	*	**.083**	**.083**	.019	**None**	99.7	.3	–0–	–0–
General Asphalt	**14.0**	2.3	**4.1**	.441	.438	d.037	d.016	d.016	.024	**	98.1	1.5	–0–	.4
International Cement	10.2	2.4	2.8	.420	.212	**.049**	**.021**	**.013**	**.044**	1.7	62.4	2.3	–0–	35.2
Johns-Manville	8.6	1.9	**4.9**	**1.03**	**.647**	.008	.009	.007	.033	1.4	76.9	.8	22.3	–0–
Otis Elevator	**17.9**	**5.7**	*	*	*	*	d.067	d.055	.027	**	81.1	.9	18.0x	–0–
U. S. Gypsum	**15.5**	**5.2**	*	*	*	*	**.033**	**.028**	**.035**	**3.9**	84.9	1.4	13.6x	–0–
Average	11.4	2.6	3.0	.584	.452	.015	.011	.009	.033	1.9	81.8	1.2	9.1	7.9

CHAIN STORES

First National Stores (3/31/35)	3.8	1.0	**9.3**	**5.41**	**6.48**	.029	.154	.136	**.066**	13.2	86.3	2.0	11.7x	–0–
Grant (W. T.) (1/31/35)	**7.4**	**1.5**	**8.2**	**2.97**	**4.48**	.031	.091	.082	**.061**	**None**	88.0	1.9	–0–	10.1
Kresge (S. S.)	4.5	**1.6**	7.9	1.49	1.49	**.070**	.105	.085	.040	8.8	78.1	3.3	1.7x	16.9
Kroger Grocery	4.4	1.2	**10.6**	**4.66**	**7.73**	.019	.088	.088	**.082**	**583.1**	98.7	1.1	.2x	–0–
Neisner Bros.	**7.3**	**2.8**	**11.6**	**4.16**	2.51	.041	**.171**	.074	.035	3.0	41.0	2.4	23.6x	33.0
Penney (J. C.)	5.4	1.2	5.1	**3.92**	*	**.073**	**.287**	**.241**	*	**26.1**	83.0	.9	16.1x	–0–
Safeway Stores	4.7	1.1	**10.8**	**8.00**	**7.26**	.012	.095	.064	**.085**	3.9	66.2	1.1	32.7x	–0–
Woolworth (F. W.)	**13.2**	**1.5**	7.1	1.54	**3.81**	**.119**	**.182**	**.182**	.024	**None**	98.4	1.1	–0–	.5
Average	5.7	1.3	7.9	2.86	3.68	.055	.156	.138	.056	21.6	87.1	1.5	6.3	5.1

CHEMICALS

Air Reduction	**8.3**	**8.3**	**9.6**	.516	.624	**.248**	**.128**	**.128**	**.053**	**None**	98.7	1.3	–0–	–0–
Allied Chemical	**8.3**	**3.8**	*	*	*	*	.086	.070	*	6.4	70.9	10.5	18.6x	–0–
Amer. Agric. Chemical (6/30/34)	**21.0**	**2.4**	*	*	*	*	.056	.056	**.068**	**None**	97.9	2.1	–0–	–0–
American Cyanamid	3.6	1.5	*	*	*	*	.063	.053	.034	6.1	71.8	11.4	3.0	13.8
Columbian Carbon	**8.1**	**2.6**	**4.4**	.382	.280	**.216**	.083	.079	.027	**None**	94.8	.8	4.4y	–0–
Commercial Solvents	**13.2**	**2.9**	*	*	*	*	**.146**	**.146**	**.058**	**None**	98.6	1.4	–0–	–0–
DuPont (E. I.)	6.5	**2.4**	*	*	*	*	**.101**	.079	**.049**	7.1	72.3	6.1	21.4x	.2
Freeport Texas	3.4	1.0	1.6	**.641**	.655	.159	**.102**	**.093**	**.047**	**19.1**	90.5	1.3	8.2x	–0–
Glidden Co. (10/31/34)	3.4	1.1	**3.8**	**1.89**	**1.88**	.037	.069	.043	.030	2.7	48.0	13.5	25.4x	13.1
Hercules Powder	**13.9**	2.2	**3.6**	**1.02**	**.853**	.089	.091	.062	**.059**	4.1	55.2	13.8	31.0x	–0–
Mathieson Alkali	2.3	1.1	*	*	*	*	.048	.043	.036	7.0	89.0	.8	10.2x	–0–
Monsanto Chemical	4.7	1.8	*	*	*	*	**.167**	**.142**	**.057**	**17.2**	83.5	1.1	10.5x	4.9
Pittsburgh Plate Glass	**7.3**	2.2	*	*	*	*	.067	.067	*	**None**	99.2	.7	.1y	–0–
Sherwin Williams (8/31/34)	**7.0**	1.8	*	*	*	*	**.110**	.074	*	4.7	65.5	1.6	32.9x	–0–
Texas Gulf Sulphur	**16.4**	1.7	1.2	.297	.669	**.408**	**.121**	**.121**	.020	**None**	99.0	1.0	–0–	–0–
Union Carbide & Carbon	6.2	1.5	*	*	*	*	.090	**.084**	.032	**18.0**	92.7	.7	2.9x	3.7
United Carbon	3.6	**2.9**	**7.5**	.426	.290	**.249**	**.106**	**.106**	.041	**15.9**	93.9	2.4	.1y	3.6
U. S. Ind. Alcohol	**7.1**	1.9	*	*	*	*	**.132**	**.132**	**None**	**None**	97.2	2.8	–0–	–0–
Average	6.7	2.2	2.9	.619	.684	.171	.097	.080	.041	9.3	79.7	5.1	13.7	1.5

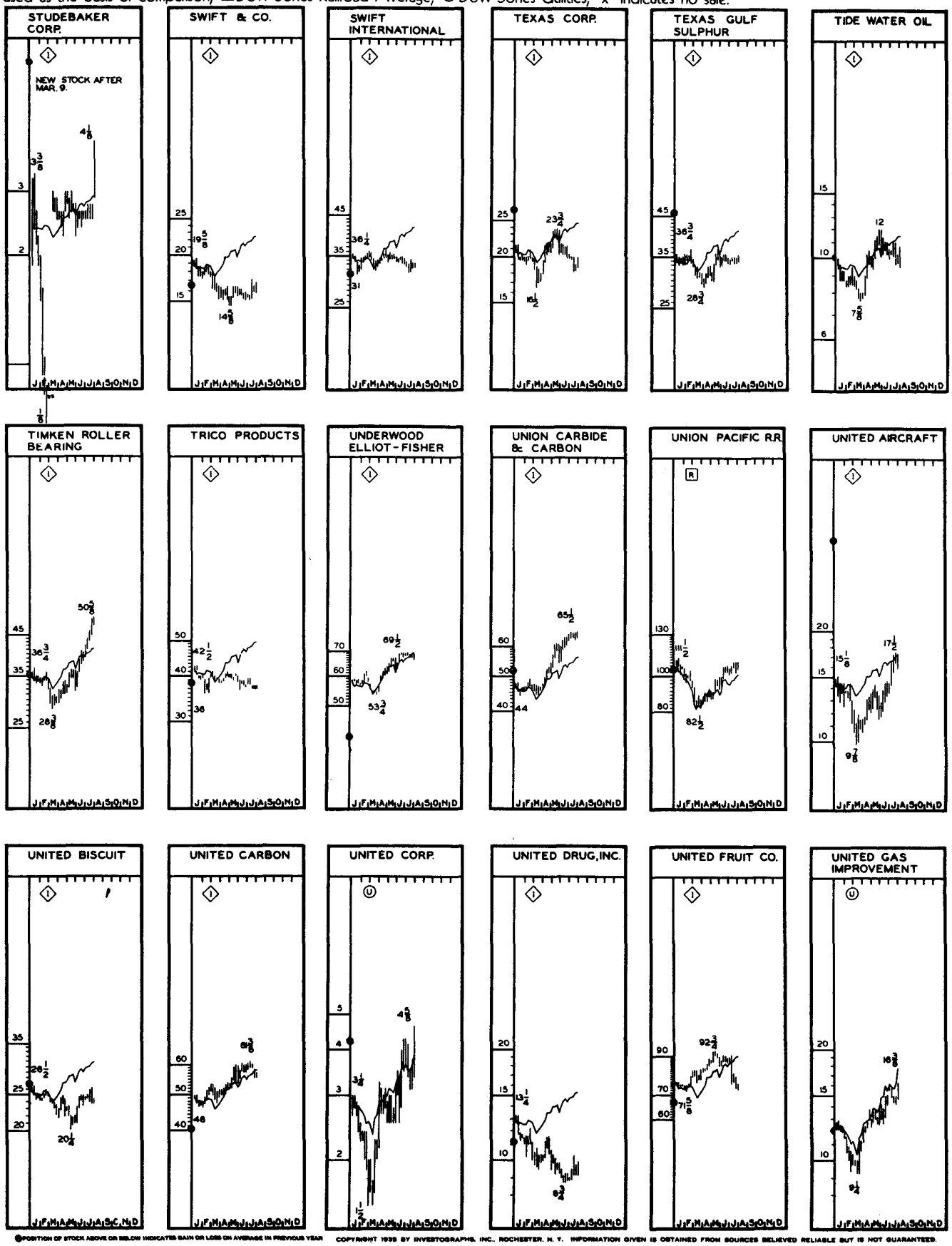
*not available **not earned d—deficit x—no minority interest y—no preferred

Explanation: (1) Number of dollars in current assets for each dollar in current liabilities. (2) Denotes dependency of working capital on inventory. (3) Shows relation of annual sales to inventory. (4) Amount in sales for each dollar of net worth. (5) Amount in sales for each dollar of plant or productive assets. (6) Amount of profit on each dollar of sales. (7) Amount of profit from each dollar of net worth. (8) Amount of profit from each dollar of capitalization. (9) Amount charged to depreciation for each dollar of plant.

Copyright 1935 by Investographs, Inc., Rochester, N. Y. Information given is obtained from sources believed reliable but is not guaranteed.

INVESTOGRAPH COMPARATIVE MARKET CHARTS THRU AUG. 3, 1935

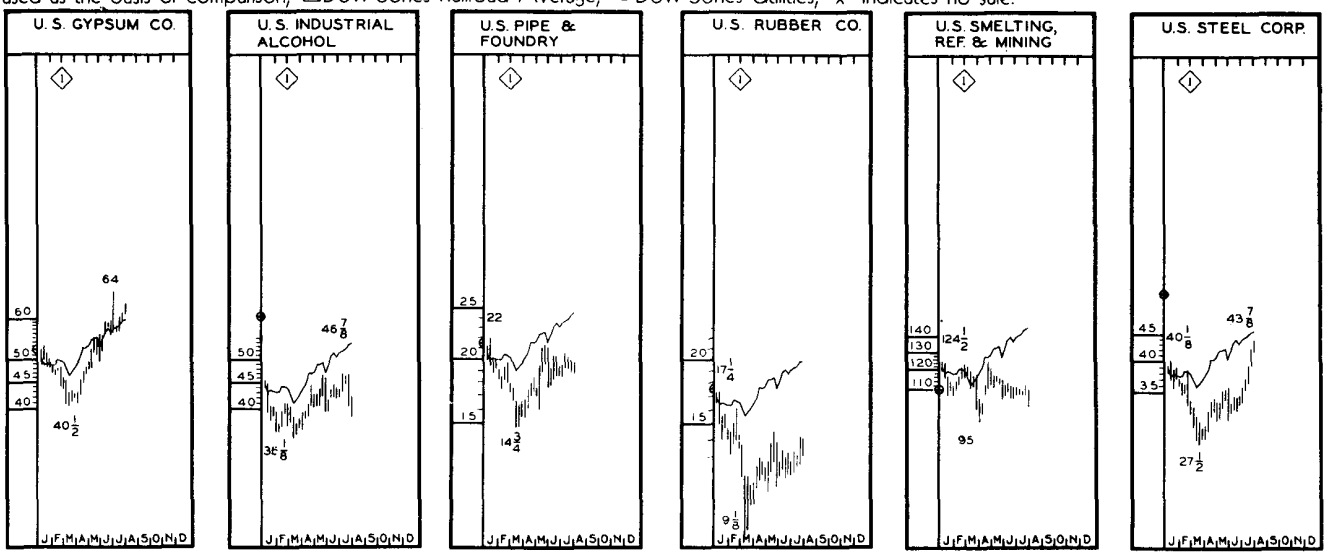
Price ranges, on a weekly basis, are shown by the vertical bar charts. The superimposed line charts show the *relative* market action of the proper Dow-Jones average for comparison. The use of a "ratio" scale makes this possible. The mark ⓘ means the Dow-Jones Industrial Average has been used as the basis of comparison, Ⓡ Dow-Jones Railroad Average, Ⓤ Dow-Jones Utilities; "x" Indicates no sale.

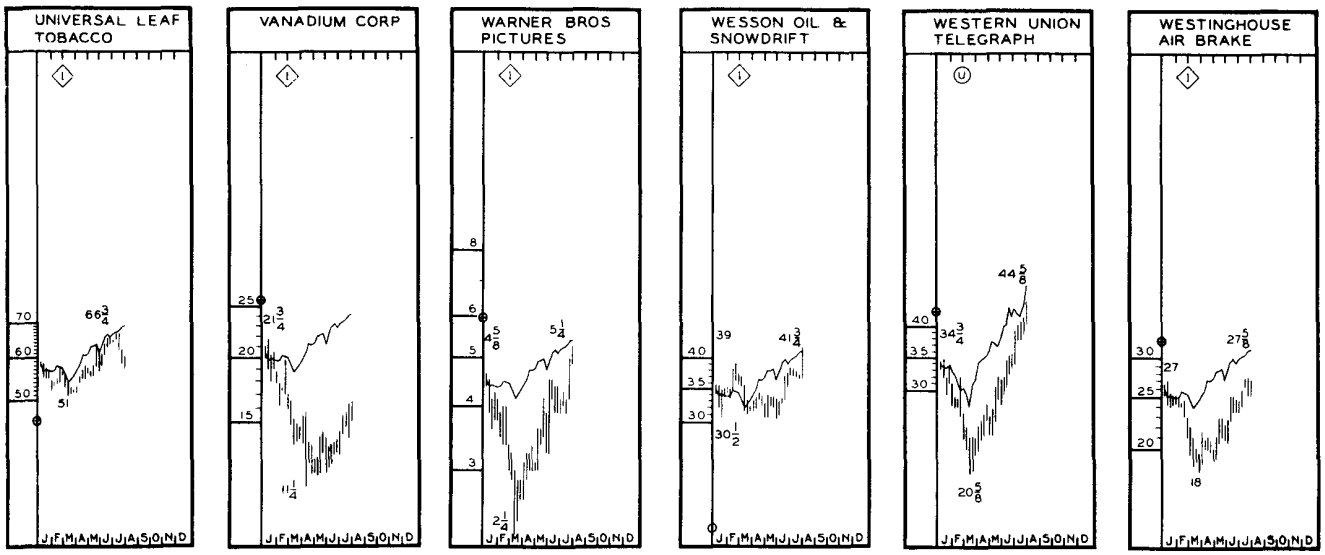


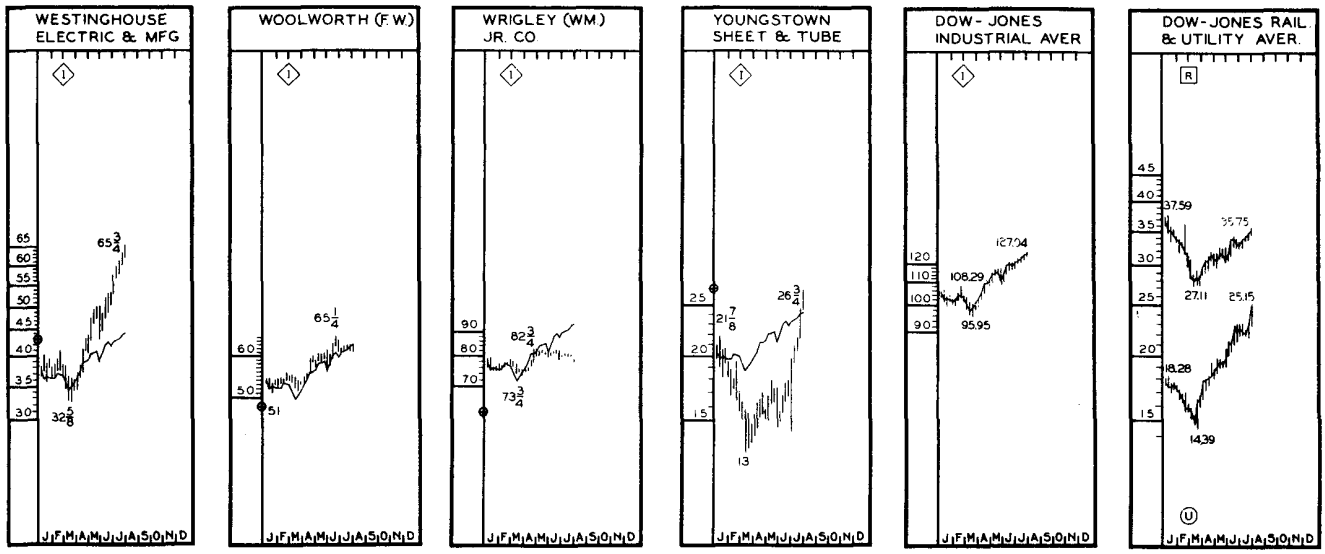
ⓘ POSITION OF STOCK ABOVE OR BELOW INDICATES GAIN OR LOSS ON AVERAGE IN PREVIOUS YEAR COPYRIGHT 1935 BY INVESTOGRAPHS, INC. ROCHESTER, N. Y. INFORMATION GIVEN IS OBTAINED FROM SOURCES BELIEVED RELIABLE BUT IS NOT GUARANTEED.

INVESTOGRAPH COMPARATIVE MARKET CHARTS THRU AUG. 3, 1935

Price ranges, on a weekly basis, are shown by the vertical bar charts. The superimposed line charts show the *relative* market action of the proper Dow-Jones average for comparison. The use of a "ratio" scale makes this possible. The mark ⓘ means the Dow-Jones Industrial Average has been used as the basis of comparison, Ⓡ Dow-Jones Railroad Average, Ⓤ Dow-Jones Utilities; "x" Indicates no sale.







ⓘ POSITION OF STOCK ABOVE OR BELOW INDICATES GAIN OR LOSS ON AVERAGE IN PREVIOUS YEAR

COPYRIGHT 1935 BY INVESTOGRAPHS, INC. ROCHESTER, N. Y. INFORMATION GIVEN IS OBTAINED FROM SOURCES BELIEVED RELIABLE BUT IS NOT GUARANTEED.

CHAPTER II

CHARTS — THE AVERAGES

REFERENCES

"Statistical Methods" (Chapter II)	Frederick C. Mills
"Statistical Analysis" (Chapter V, Appendix E)	Edmund E. Day
"Graphic Charts in Business"	A.C. Haskell
"How to Make and Use Graphic Charts"	A.C. Haskell
"The Ratio Chart in Business"	P.A. Bivins
"Graphical Methods"	W.C. Marshall
"Graphic Methods for Presenting Facts"	W.C. Brinston
"Charts and Graphs"	K. Karsten

In the previous Chapter, it was noted that technical studies are founded upon the observation of stock price trends.

Charts or graphs are the best means man has so far achieved to record the history of stock prices so that they may be quickly and easily visualized. They assist and aid the memory in recalling important events of the past which have significant bearing upon the present and future. Their primary function is to provide accurate factual data. Just as the fundamental student is accustomed to turning to constantly revised reference books which contain the previous records of assets and liabilities, income and expense, sales and working capital, and other pertinent fundamental facts, the technical student consults his stock price charts which record the final judgment of all interested persons — namely the price trend itself.

Used with sober judgment, charts can be of the greatest benefit to the stock trader. In the hands of an untrained amateur, they are a very dangerous instrument, causing many financial fatalities. Used in place of a crystal ball, they are to be scorned. Employed as an adjunct, as one means among others to form sound judgment by being alert to all the facts, they can be extremely useful and valuable.

Unfortunately, charts are in bad repute in some parts of the financial circle, because from time to time a vast herd of stock market dabblers have tried to employ crude charts, as the mystic casts his spell.

There are many kinds of charts, some of which are absolutely useless. Others, specifically designed for the purpose of studying stock price trends, are of unquestionable value in providing part of a broad general understanding.

The severest critics of charts and chartists are usually those who, learned in other respects, have not taken the time or made the effort to learn the developments which have been taking place in the field of technical studies by means of charts.

In this Chapter, we will try to dispose of the mechanical details concerning the preparation and the use of charts which are best suited to the study of stock price trends. As we will be dealing with stock market averages, we will also look into this necessary subject. Then, in following Chapters, we will proceed to study the important aspects of the technical approach.

Charts are so essentially a part of technical studies that their use cannot be avoided. The stock trader who feels that they are an unnecessary nuisance is fooling only himself.

Records Are Necessary in Any Well-Run Business

The business of trading in stocks is no different from any other, in that it requires a knowledge of what is going on, which in turn necessitates records. This is particularly true when one considers the vast volume of trade which is conducted on the floor of the Stock Exchange each day. Because the market is no simple affair, a considerable selection of information is essential. To study all that occurs is an impossible and useless procedure.

The Need For Records

One of the most important tasks with which the technical student is confronted is to develop a system of record-keeping. There are two reasons for this:

1. Every principle employed in technical analysis has been developed from the study of the *records* of past market action.

2. Practically all technical indications require time for their development; hence in interpreting the current market the student must have before him the *record* of the relevant past.

Some stock market students, it is true, endeavor to dispense with formal records, and keep the details of the price movement in their heads, but the numerous figures with which we deal, and the need for their visual presentation rule out this type of mental bookkeeping at once. *In the method of technical analysis we are to study, an orderly set of records scrupulously kept up to date is a prerequisite.*

Two Types of Records Employed

The records employed in technical work are of two types. These are: